SignalRx: finding what pharma filings don't say on their own

Autonomous financial research agent · **Track D: Pharma & Healthcare** · Finding the Next Signal, ATMOS 2026

TeamName
Member 1 · Member 2 · Member 3 · Member 4
MVP: github.com//signalrx

Idea. In pharma and healthcare the important story usually sits *between* documents, not inside one: an FDA inspection at a contract plant the company doesn't own, a government payor's billing order, a loss-of-exclusivity date, a partner's filing, a peer's results for the same quarter. SignalRx builds a dated **evidence ledger** across these independent source families, lets the numbers say *where* something moved, sends an LLM investigator to find *why*, and then tries to **kill its own hypothesis** before reporting anything. A summariser reads one source; SignalRx reports only findings that need at least two independent families and calculations no single source states.

MVP PILOT RESULTS (WORKING CODE, REPLAYABLE; ALL PASS THE OUTPUT VERIFIER)

Company	Discovery the agent produced (abridged)	Brief	Strength
Dr. Reddy's	North America is down to **27.4%** of revenue (Q1FY27), **below even FY21's 37.8%**, while ex-NA revenue grew **+23.5%** in FY26. The lenalidomide cliff is already in the numbers, so **semaglutide is now the swing factor**, and it has slipped twice in 8 months: a Health Canada notice (Oct-25), then an out-of-spec API (Jul-26), leading to a ₹240 cr write-down (**54% of Q1 profit**) and a pen plan cut from 12M to 6–7M.	D1/D3	Strong
Cipla	The sharpest US decline came from **a plant Cipla doesn't own**: the exclusive lanreotide maker (Pharmathen, Greece) closed an FDA inspection with 9 observations on 21-Nov-25. NA then fell **28% QoQ** ($233M→$167M) in a quarter that **closed 14 days before Cipla disclosed it**. The skeptic shows lanreotide explains only **~51%** of the step, and the $1bn exit-rate target needs **+54%** with zero lanreotide.	D2	Strong
Max Healthcare	The H2-FY26 **ARPOB stall was a payor-policy shock, not demand**. After CGHS drug-billing rules, Max stopped high-value chemo for institutional patients, and oncology fell from **26% to 22%** of inpatient revenue. Reported growth (16%) understates the core (20%), ~1.2pp of the Q1 ARPOB rebound is the CGHS tariff revision, and the "normalisation" date is just the base effect lapping. *Secondary: the bed target needs 3.3–3.7× the current build pace.*	D2	Moderate

WORKFLOW: COMPANY → PLAN → EVIDENCE → DISCOVERY

resolve → harvest → verify inputs → profile → scan → plan → investigate → expand 3rd parties → hypothesise → skeptic → adversarial search → score → write → verify output → render

Blue = deterministic code; orange = LLM. The LLM **proposes**; code **disposes** (it computes, checks and scores).

1. **Harvest**: 5–10 years of statements and quarterly results, Reg-30 filings, FDA recalls/approvals, dated headlines.
2. **Scan**: where the numbers moved: mix shifts, segment step-downs, reported-vs-adjusted growth gaps, ARPOB stalls, announced-vs-commissioned capacity.
3. **Plan**: ranks the Track D angle library (D1 model drift, D2 regulatory-before-coverage, D3 geography, plus third-party plants, payor policy, capacity, narrative gap) by archetype fit × anomalies, with one open LLM-proposed slot.
4. **Investigate**: a Claude tool loop per angle (search, fetch, PDF search, FDA/NSE/news). Each record needs a verbatim quote from a fetched document **with exactly matching numbers**, or it is rejected.
5. **Expand**: each third party found (contract maker, supplier, payor) gets its own regulator probe. This is how the Cipla finding is reached.
6. **Hypothesise → challenge → score → write**, then the verifier re-traces every number in the prose.

GENERALISES ACROSS THE TRACK, WITH NO COMPANY TUNING

- **Universe from NSE index files** at run time (today Sun, Cipla, Dr. Reddy's, Apollo, Max): reshuffles need no code.
- **Archetype from the company's own text and segments** (formulations / API-CDMO / hospital / diagnostics) routes to the right KPIs. The pilots cover two archetypes on one code path, which the tie-break pool (Divi's, Lupin, Fortis, Syngene...) shares.
- **Sector knowledge pack, not company knowledge**: FDA/Health Canada/EU escalation ladders, a CGHS/NPPA/Sec-232/BIOSECURE watchlist, an LOE calendar and a controlled metric vocabulary. A test fails the build if any company name appears in code.

TRIANGULATION AND CONTRADICTION DETECTION

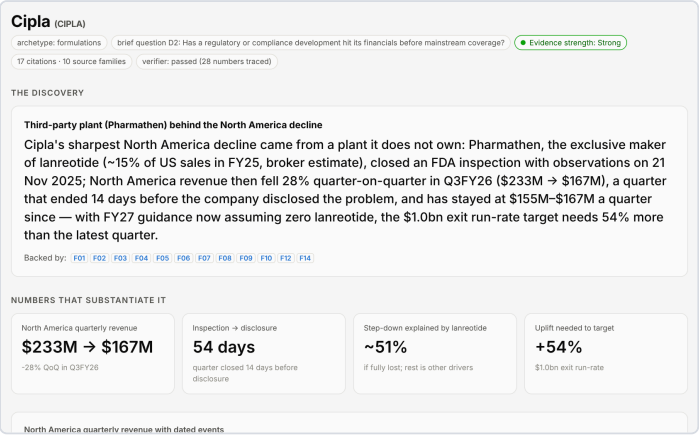
- **Source families** (company · each regulator · each counterparty · each publisher): a discovery needs ≥2, and the strength grade counts them.
- **Recompute, don't trust**: reported growth is recomputed from filed levels. In the pilot, one outlet printed Dr. Reddy's Q1 at **+6%** and NA at **+35%**; the filings give **−5.6%** and **−35.4%**, so both were dropped and logged.
- **Hierarchy**: regulator = filing > transcript/counterparty > broker/press > aggregator > blog. Overruled figures go to an audit log; derived values carry tolerance bands. Peers are compared for the same quarter.

SELF-FALSIFICATION AND TEMPORAL REASONING

The skeptic is code that **re-runs the numbers** for each alternative: sector-wide · acquired · one-off (adds it back) · temporary (dated fix becomes a falsifier) · second driver *with timing* · breadth · low-margin · already-in-headlines · data conflicts. Live runs also send an **adversarial investigator** after the leading hypothesis. Each event carries a date and a precision; the agent measures regulator → disclosure → headline → results lags. From the pilots:

- Cipla: the lenalidomide LOE (31-Jan-26) lands *after* the step quarter, so the finding was resized to "~51% explained".
- Dr. Reddy's: broad ex-NA growth reframed "concentrated replacement" into "semaglutide is the swing factor".
- Max: "normalisation by Q3–Q4 FY27" falls 13 months after the change, which the agent flags as base-effect lapping, not recovery.

OUTPUT: INTERACTIVE DASHBOARD PLUS PDF RESEARCH NOTE (SAME CONTENT)



Top to bottom: the discovery (one sentence, citation chips), key-number tiles, a chart with dated event markers, why it matters, the strongest counterargument, falsification conditions, a timeline, evidence strength with reasons, the skeptic table, the full evidence trail (stance, family, passage, link; filterable), every calculation with formula and inputs, the hypothesis board including rejected ideas, the audit log and the agent trace. **Every claim is one click from its passage and URL.** The PDF is the same page printed to A4.

EVIDENCE STRENGTH RUBRIC (SHOWN WITH ITS REASONS)

- **Strong**: ≥4–6 independent families incl. a primary filing, most skeptic tests survived, no refutation, all conflicts resolved.
- **Moderate**: ≥3 families incl. a primary filing, but key attribution rests on management commentary or the press.
- **Weak**: thin or single-family support, or a refuting test. A weak finding is still shown, with that label.

The ranking score is 0.35 novelty + 0.30 evidence + 0.20 materiality + 0.15 robustness. Novelty rewards non-company families, a third-party record that *leads* the company's disclosure, and derived calculations; it is penalised when headlines already carry the core terms.

READINESS ACROSS THE TRACK: LEADS FOR THE OTHER TWO NIFTY 50 NAMES

From the same research protocol; not yet run end to end. These are the angles the planner would pick first.

Sun Pharma	D1 + policy: innovative medicines reached 21.9% of sales (+12.8% to $351M) while US formulations fell 9.7% to $427M (Q1FY27), with three US-facing plants under FDA warning letters. Test whether the specialty pivot raises exposure to the 100% Sec-232 tariff on patented imports (Sep-26).
Apollo Hospitals	Capacity gap: the FY26 plan of six facilities and ~1,400 beds narrowed to four hospitals, with ~855 beds commissioned by May-26 against a 4,400-bed five-year target. Narrative gap: HealthCo's first profitable year vs the slipping Keimed merger timeline.

TOOLS AND STACK

Layer	Choice
Reasoning model	Claude Opus 5.5 (`claude-opus-5-5`): adaptive thinking, JSON-schema structured outputs, strict tool schemas, server-side refusal fallbacks. Swappable behind one wrapper.
Agent loop	Our own manual tool loop (handles `pause_turn`; parallel tool results). No framework lock-in, and every step is logged to the trace.
Search / retrieval	Anthropic server-side web search; our own fetcher with an on-disk cache; HTML/PDF to text (BeautifulSoup, pypdf); BM25 inside long filings; fuzzy quote grounding with exact numbers.
Public data	NSE/BSE Reg-30 filings · results releases, investor decks, transcripts, annual reports · SEC EDGAR 6-K/20-F for ADR issuers · Screener statements · openFDA recalls & Drugs@FDA approvals · FDA Data Dashboard, warning letters, import alerts, Form 483 FOIA · Health Canada NOC · CGHS/NPPA orders · ClinicalTrials.gov v2 · news RSS (first-coverage dating). No paid data.
Analytics	Python calculators (shares, growth, implied priors, required run-rate, decomposition); every derived number is a recorded calculation with its inputs.
Output	Jinja2 + inline SVG (no JS libraries; works offline, light and dark), Chromium to PDF; `run.json` holds the full ledger and trace.
Quality	25 pytest tests: calculators, verifier, grounding (rejects a quote with a changed number), parsers, the LLM loop against a fake client, and the generality guard.

BUILD PLAN TO THE FINALS (25 OCT)

16–19 Oct	**Done:** full replay pipeline, three pilot discoveries across two archetypes, dashboard + PDF, 25 tests; Round 1 submission.
20–21 Oct	Live connectors smoke-tested on an open network (`scripts/smoke_live.py`); live runs on all five NIFTY 50 track companies, each frozen into a replayable bundle.
22 Oct	Tie-break pool dry runs (Divi's, Lupin, Zydus, Torrent, Aurobindo, Fortis, Narayana, Syngene). Track verifier pass rate, families per discovery, time to discovery (target <12 min), and false-claim rate from a manual audit of 20 citations.
23–24 Oct	Harden against what the dry runs reveal (parser drift, new angles such as specialty-pivot and own-site OAI templates); rehearse the 10-minute demo: a live run on the judges' pick with a replay fallback, plus Q&A.
25 Oct	Finals: live run on the company the judges choose; dashboard walk-through from claim to source; counterargument and falsification on screen.

Limits we state up front. Replay bundles were collected with a search tool, so passages are search-result snippets; live runs replace them with full-document quotes. Third-party facts (FDA 483 details) currently arrive via the press; the live regulator probes add the primary record. No investment recommendation is made.